Softwaregestützte Modellbildung und Simulation: Exercise 2

Principal Component Analysis (PCA)

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Principal Component Analysis (PCA) is an application of Singular Value Decomposition (SVD) ...

The data set is ...

1. Load the data into a single NumPy array, using `numpy.loadtxt()`. Mathematically, we will call this $\mathbf{X}$. Note that the rows of $\mathbf{X}$ are the separate *samples* (measurements), while its columns are the variables: *sepal length*, *sepal width*, *petal length* and *petal width*.

2. Center the data by dividing each column of $\mathbf{x}$ by its mean:

$$\mathbf{Y} = \mathbf{X} - \overline{\mathbf{X}}.$$

3. Normalize each column of $\mathbf{Y}$ by dividing it by its standard deviation σ:

$$\mathbf{B} = \frac{1}{\sigma}\mathbf{Y}.$$

4. Apply the SVD method on the matrix $\mathbf{B}$:

```python
U, S, VT = numpy.linalg.SVD(B, full_matrices=False)
```

(where `U, S, VT` are, respectively, the matrices $\mathbf{U}$, $\mathbf{\Sigma}$, and $\mathbf{V}^\intercal$ in the SVD of $\mathbf{B}$)

5. In the terms of PCA, the columns of $\mathbf{U}$ are called the *Principal Components* (PC) of $\mathbf{B}$. Their corresponding singular values are the diagonal elements of $\mathbf{\Sigma}$, and

they are ordered by absolute value in descending order. The variance in PC_i is given by

$$\mathrm{Var}_i = \frac{\sigma_i^2}{N-1},$$

where N is the number of samples. Plot the variance for each PC in descending order.

6. If we normalize the variance by the sum of all variance terms, we get the variance in each PC in terms of the percentage of total variance of the data. Calculate these percentages and adjust the previous plot accordingly.

7. The cumulative sum of the normalized variances up to (and including) the i-th PC gives us an representation of how much of the variance in the data is described by the first i PCs. Plot the cumulative sums against the components (i.e. cumulative sum vs. i). Does this value converge to a limit? If so - how can you explain this limit?

8. ...